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Role of symmetry in self-healing of singular beams

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Abstract

The phase discontinuities in the wavefront smooth out during propagation. This smoothing out phenomenon by diffraction in some cases leads to healing of the wavefront thereby restoring the original field distribution, even when the wavefront is subjected to obstructions. We, for the first time, report that the symmetry of the obstruction plays a crucial role in the healing process of an optical beam. Singular beams are seen to heal on facing an obstruction that is symmetric about the optical axis of the beam. It is found that scalar, as well as vector vortices, do not heal under asymmetric (non-symmetric) perturbations. Vector field singularities (V-points) split into multiple ellipse-field singularities (C-points) under asymmetric obstructions based on the index, and helicity conservation rules. Stokes parameter S_3 and Stokes phase distributions (Φ_{12}) are also examined to observe the split of V-points into C-points clearly. The perturbation of the V-points with asymmetric obstacles can be used as a generation technique for producing C-points from V-points. All the above claims are well supported in the article with the appropriate experimental results

1 Introduction

Diffraction of optical beams tends to smooth discontinuities in the wavefronts. The Huygens' spherical waves which play a crucial role in diffraction and wave propagation, have a propagation vector that is diverging in nature and have no transverse component to the general propagation direction. Any transverse components of the propagation vector field distribution that arises due to local tilts in the wavefront are mostly irrotational. Singular beams are special types of beams in which the transverse components of the propagation vector field distribution are solenoidal [1]. By definition, singularity [2] refers to a point where the function (F) is undefined and the line integral of the gradient of function enclosing the singular point is non-zero, i.e., $\oint \nabla F \cdot ds \neq 0$, where ds is the line element representing the

path of integration. This implies that its neighborhood is characterized by high gradients and the vector field ∇F is solenoidal. Phase singularities refer to wavefront dislocations and the structure of the wavefront is characterized by a helical ramp. For polarization singularities, the polarization azimuth variation has a dislocation.

The self-healing observed in singular beams is due to the transverse components of the propagation vector field distribution. However, there are cases when self-healing is observed for beams with no rotational components of propagation vector, for instance, a Gaussian beam [3]. A Gaussian beam is not a singular beam, but a singularity can be implanted in the beam waist of the Gaussian beam like in special modes of Laguerre–Gaussian (LG) beams, higher order Bessel beams, Airy and Mathieu beams. Due to the fact that the Fourier transform of a Gaussian beam is also a Gaussian [4], it has propagation invariant field distribution. Another example of a self-Fourier transform beam is a vortex beam [5, 6] which has also shown self-healing property [7]. Symmetry aspects of the beam and the obstruction in the healing process have, to the best of our knowledge, not been reported in the literature so far. The main objective of this article is to study the role of symmetry in the healing process of the singular beams.

It was first demonstrated by Vasnetsov et al. that an obstructed singular beam could reconstruct itself from its remaining part that does not contain the vortex [7]. This

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regeneration is attributed to the redistribution of the residual orbital angular momentum in the beam's cross-section. It was observed that as the beam propagates, the region of obstruction recovers on its own without any external perturbation. This self-reconstruction of the obstructed beam on its propagation is termed self-healing. This healing phenomenon has been noticed in the intensity [8–11], phase [7], and polarization distributions [12, 13] of different optical beams. Moreover, the coherence properties of optical beams are observed to heal during propagation [14]. There are many explanations for this interesting healing phenomenon for different types of optical beams [15]. It is claimed that any optical beam that can be represented in the angular momentum representation can self-heal [3]. Self-healing in some beams is attributed to their non-diffractive nature, while in others, it is explained using caustic, ray, wave, and ray-wave theory [15].

Singular beams perturbed with rotationally symmetric obstructions about the optical axis (singular point), show good healing capacity due to the rotational symmetry in the internal energy flow pattern [16]. However, there is still a lacuna in the complete study of self-healing. Scalar vortices are reported to self-heal for various cases of perturbations [7, 9, 10, 17–19]. Vector vortices are also reported to heal under different types of perturbations [13], but this is not always the case. For both symmetric as well as asymmetric perturbations, it is only intensity that seems to self-heal [20, 21]. A clearer picture is visible when the polarization parameter is also taken into account. In this paper, the effect of symmetrical and asymmetrical perturbations in the healing process of optical singularities is studied. A schematic diagram showing the symmetric and asymmetric perturbation of a vortex beam is depicted in Fig. 1. It is observed that symmetry plays an important role in the healing process. Both scalar and vector vortices are seen to heal under symmetrical amplitude obstruction. Partial healing in the intensity distribution is observed in scalar vortices while vector vortices split into lower order singularities under an asymmetric obstruction. This article is organized as follows: in Sect. 2, the role of symmetry is studied in the healing process of scalar vortices. In Sect. 3, vector vortices, specifically V-points, and their response to both symmetric and asymmetric obstructions are discussed. Experimental verification for these results is provided in Sect. 4, followed by a conclusion in Sect. 5.

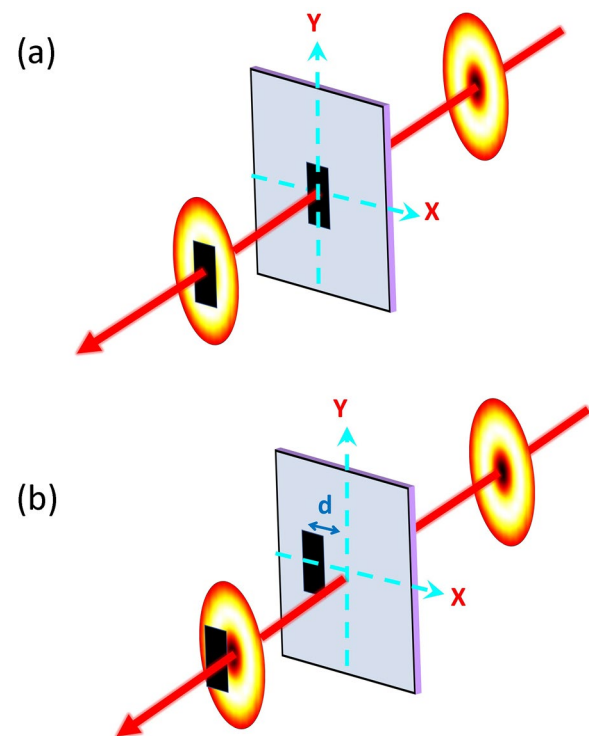


Fig. 1 Schematic diagram showing symmetric and asymmetric blocking of vortex beam through the rectangular obstruction

2 Role of symmetry in the healing of scalar vortices

Scalar vortices [22, 23] have a helical wavefront and are defined as beams with an azimuthal phase factor. A vortex beam of topological charge l can be mathematically expressed as

$$\psi = r^{|l|} \exp(i\Theta) \times \exp\left(\frac{-r^2}{w_0^2}\right), \quad (1)$$

where r is the radial coordinate, w_0 is the beam waist and $\Theta = l\theta$, where θ is the azimuthal angle. The vortex beam used in the above expression corresponds to an r-type vortex [23–25] with a Gaussian envelope. The phase cannot be defined at the location of the vortex, and hence, is also termed phase singularity. The topological charge l of these vortices is defined as [26]

$$l = \frac{1}{2\pi} \oint \nabla \Theta \cdot ds, \quad (2)$$

where ds is the line element that represents the path of integration. The effect of the symmetry of obstruction on these vortices can be studied through the Fraunhofer diffraction pattern of the obstructed field. Let $T(r, \theta)$ be the

transmittance function of the obstacle. The obstructed field in the far-field (ψ_f) can be written as

$$\psi_f = \mathcal{F}[\psi \times T] = \mathcal{F}[\psi] \otimes \mathcal{F}[T], \quad (3)$$

where ψ is the initial unobstructed vortex beam and $\mathcal{F}$ represents the Fourier transform. The Fourier transform of a scalar vortex beam of unit amplitude with topological charge l can be written as [5, 6]

$$\mathcal{F}[\psi] = \frac{|l|(-i)^{|l|} e^{il\phi}}{2\pi\rho^2} \otimes \pi w_0^2 \exp(-\pi^2 w_0^2 \rho^2). \quad (4)$$

In this equation, $\rho = \sqrt{f_x^2 + f_y^2}$ and $\phi = \tan^{-1}(\frac{f_y}{f_x})$, where (f_x, f_y) are coordinates of Fourier space. From Eq. 3, the field ψ_f can be written as

$$\begin{aligned} \psi_f &= \mathcal{F}[\psi \times T] \\ &= \frac{|l|(-i)^{|l|} e^{il\phi}}{2\pi\rho^2} \otimes \pi w_0^2 \exp(-\pi^2 w_0^2 \rho^2) \otimes \mathcal{F}[T] \end{aligned} \quad (5)$$

using the convolution theorem. For simplicity in mathematical analysis, we assume a circular obstacle with radius a obstructing a beam of radius b where $a < b$. The transmittance function of the circular obstacle is given by

$$T(r) = \begin{cases} 0, & \text{if } r \leq a \\ 1, & \text{otherwise.} \end{cases} \quad (6)$$

Therefore,

$$\begin{aligned} \psi_f &= \frac{|l|(-i)^{|l|} e^{il\phi}}{2\pi\rho^2} \otimes \pi w_0^2 \exp(-\pi^2 w_0^2 \rho^2) \\ &\otimes \left(\frac{bJ_1(2\pi b\rho)}{\rho} - \frac{aJ_1(2\pi a\rho)}{\rho} \right). \end{aligned} \quad (7)$$

Now, we shift the circular obstacle by a distance d on the x -axis from the center so as to make the asymmetric perturbation. The modified field due to this asymmetry can now be written as

$$\begin{aligned} \psi_f &= \frac{|l|(-i)^{|l|} e^{il\phi}}{2\pi\rho^2} \otimes \pi w_0^2 \exp(-\pi^2 w_0^2 \rho^2) \\ &\otimes \left(\frac{bJ_1(2\pi b\rho)}{\rho} - \frac{aJ_1(2\pi a\rho)}{\rho} \times e^{(-i2\pi f_x d)} \right). \end{aligned} \quad (8)$$

In Eqs. (7 and 8), the field distributions are expressed in the form of convolution. To have an intuitive understanding, we have investigated the outcome of the convolution operation between symmetrical and asymmetrical functions. This exercise is useful, since analytical solutions in closed form may not be possible for certain types of obstacles represented by T . A circular aperture representing the input vortex beam having a circular symmetry (Fig. 2 A) is convolved with different symmetrical and asymmetrical obstacles (Fig. 2 B). It can

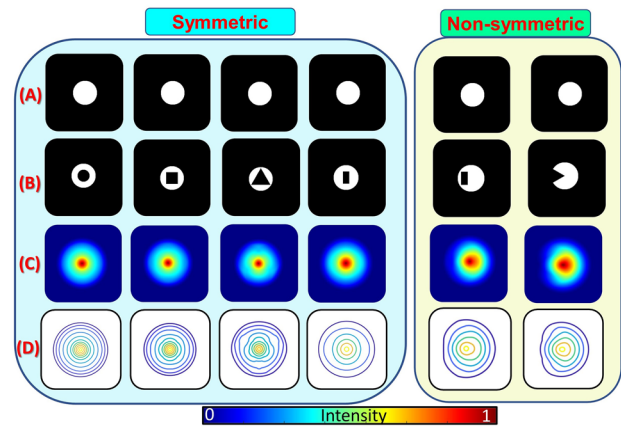


Fig. 2 Convolution of a circular symmetric function is shown with different types of symmetrical and asymmetrical obstructions. **A** Circular symmetric function, **B** symmetric and non-symmetric obstructions, **C** the resultant convolution patterns, and **D** the amplitude contour of the convolution pattern

be observed from the convolution pattern (Fig. 2 C) that the non-symmetrical obstacles disturb the circular symmetry of the beam. This leads to the disturbance in the amplitude contour of the resultant convolution pattern as illustrated in Fig. 2 D.

Obstacles of different symmetries are used for our study. The rectangular obstacle has C_2 symmetry, the triangular obstacle has C_3 , and the square obstacle has C_4 symmetry. Here, C_n symmetry refers to a n^{th} degree symmetry about an axis. This is Schoenflies notation, where point groups corresponding to one symmetry axis are denoted by C , which is accompanied by a numerical subscript n indicating the fold of the symmetry axis. The letter C is the initial letter of the word that means ‘*Cyklisch*’ (circular) [27]. This implies that the rotation of $2\pi/n$ about the axis yields the object with the same and unique conformation. It is observed from the symmetrical perturbations that the degree of healing is proportional to the degree of symmetry. This follows that the degree of healing is comparatively less for the rectangular and triangular than for the square obstacle (refer to Fig. 2). The circular obstacle has an infinite symmetry and therefore, the degree of healing is maximum in this case.

Further, the degree of healing depends on the size of the obstruction [13]. The above convolution studies help to analyze the healing phenomenon in optical vortices. The effect of symmetrical and non-symmetrical obstructions on the scalar vortices is depicted in Fig. 3. It can be seen that when the scalar vortex is obstructed by a symmetrical obstacle (triangle), the beam almost gains its circular symmetry in the far-field. The vortex structure gets disturbed initially after the perturbation but its structure gets healed with propagation. Basically, the rotational symmetry of the

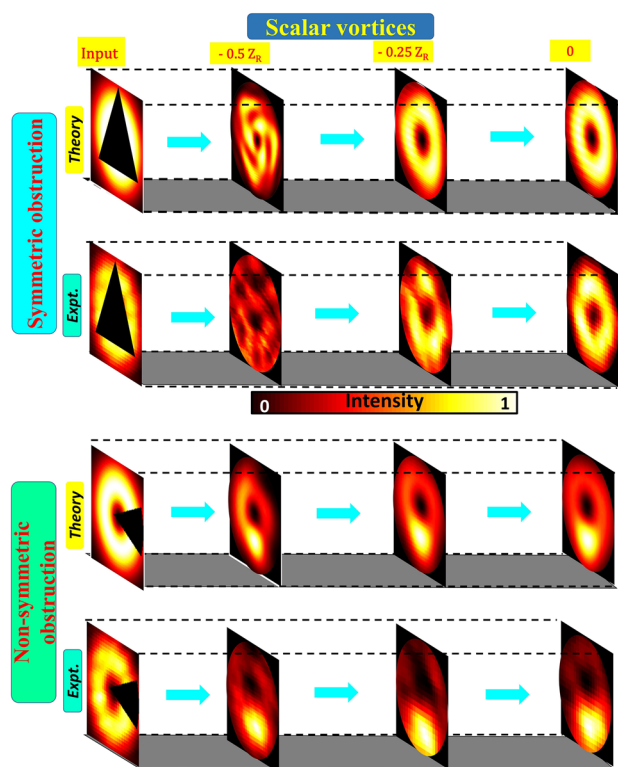


Fig. 3 Effect of symmetric and non-symmetric obstruction on the intensity healing of the input scalar vortex beam of topological charge $l = 1$. Z_R is the Rayleigh length

azimuthal flow of the Poynting vector is not disturbed in the process. On the other hand, the case is different when the scalar vortex is perturbed by a sectorial obstacle [28] that is non-symmetric in nature. The core of the vortex beam gets shifted due to this asymmetric perturbation. Hence, the scalar vortex beam is not able to regain its initial structure after a non-symmetric perturbation. Instead, only a partial amplitude reconstruction with a shifted center is observed in the far-field.

3 Role of symmetry in the healing of vector vortices

Vector vortices are points/lines of dislocations in inhomogeneously polarized beams. At these dislocations, some polarization parameters are indeterminate [29–32]. Depending on the neighborhood polarization states surrounding these vortex points/lines, they are divided into two categories: vector field polarization vortices (V-points) and ellipse-field polarization vortices (C-points). Optical beams endowed with vector vortices are shown to heal under various obstructions. In this section, we discuss the effect of symmetrical as well as

asymmetrical perturbations in the healing process of V-point beams.

These vector vortices can be represented as the superposition of two scalar optical vortex beams as

$$|\psi_{l_1, l_2}\rangle = \left(\alpha_R^{l_1} e^{il_1\theta} |R\rangle + \alpha_L^{l_2} e^{il_2\theta} |L\rangle \right) \times \exp\left(\frac{-r^2}{w_0^2}\right). \quad (9)$$

Here, $|R\rangle$ and $|L\rangle$ are circular polarization basis states that can be expressed as $\frac{1}{\sqrt{2}}[1, i]^T$ and $\frac{1}{\sqrt{2}}[1, -i]^T$, respectively. $\alpha_R^{l_1}$ and $\alpha_L^{l_2}$ are the complex coefficients that consist of amplitude and initial phase distributions. C-points are obtained when $|l_1| \neq |l_2|$, i.e., superposition of scalar vortices with unequal OAM content, while V-points are obtained when $l_1 = -l_2$, i.e., superposition of equal and opposite charged scalar vortices. V-points and C-points are characterized using topological indices namely, Poincaré–Hopf index (η) and C-point index (I_C), respectively. These indices are defined as $\frac{1}{2\pi} \oint \nabla \gamma \cdot ds$, where γ is polarization azimuth and ds is the path of integration around the singular point. Vector vortices appear as points of phase vortices in the Stokes phase map, $S_{12} = S_1 + iS_2$ [32]. Here, S_i represents the normalized Stokes parameter. The Stokes phase distribution of this S_{12} field is given by Φ_{12} . The value of S_3 is zero for the V-points since all the polarization states are linear, whereas at C-points, S_3 value is ± 1 .

3.1 Symmetrical obstruction

The role of symmetry in the healing of V-points is studied on similar lines to the scalar vortices. Since, a V-point is a superposition of scalar vortices with equal and opposite charges ($l_1 = -l_2$), in orthogonal polarization basis, the vector vortex in the far-field $\psi_f(v)$ obstructed by a circularly symmetric obstacle may be written as

$$\begin{aligned} \psi_f(v) &= \mathcal{F}[\psi_{l_1, l_2} \times T] \\ &= \frac{|l|(-i)^{|l|}}{2\pi\rho^2} (\beta_R^l e^{il\phi} |R\rangle + \beta_L^{-l} e^{-il\phi} |L\rangle) \\ &\quad \otimes \pi w_0^2 \exp(-\pi^2 w_0^2 \rho^2) \otimes \left(\frac{bJ_1(2\pi b\rho)}{\rho} - \frac{aJ_1(2\pi a\rho)}{\rho} \right). \end{aligned} \quad (10)$$

Here, β_R^l and β_L^{-l} represent complex coefficients in Fourier plane. In this case, the convolution of each scalar vortex component with a circular obstacle leads to the reconstruction of the vortex structure in individual components as discussed in the previous section. Due to the symmetry, the core of each component beam is not shifted. Each orthogonal component of the beam tries to recover itself at the far-field and superposes to obtain the resultant beam.

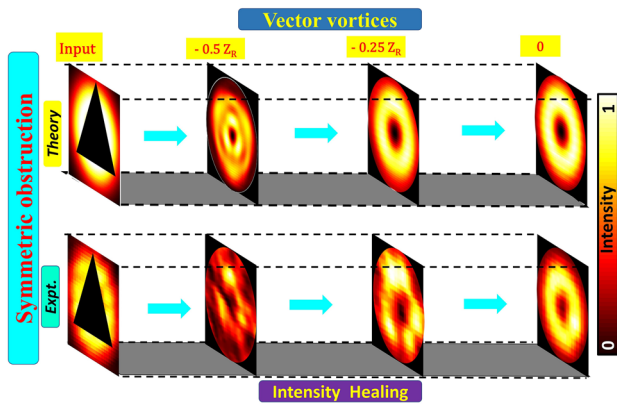


Fig. 4 Healing of the intensity distribution of the input vector vortex beam of index $\eta = 1$ after symmetric triangular obstruction

In Fig. 4, a centrally obstructing triangular obstacle is considered to study the case of symmetric obstruction of a radially polarized beam. This symmetrically obstructed vector vortex beam is seen to reconstruct its intensity in the far-field. This reconstruction or self-healing of vector vortex beams is seen with different types of obstacles having varying degrees of symmetry [13].

3.2 Asymmetrical obstruction

A totally different and interesting scenario is observed for asymmetric perturbation. For a V-point with index η , the field asymmetrically obstructed by the circular obstacle can be written as

$$\begin{aligned} \psi_f(v) &= \mathcal{F}[\psi_{l_1, l_2} \times T] \\ &= \frac{|l|(-i)^{|l|}}{2\pi\rho^2} (\beta_R^l e^{il\phi} |R\rangle + \beta_L^{-l} e^{-il\phi} |L\rangle) \\ &\quad \otimes \pi w_0^2 \exp(-\pi^2 w_0^2 \rho^2) \\ &\quad \otimes \left(\frac{bJ_1(2\pi b\rho)}{\rho} - \frac{aJ_1(2\pi a\rho)}{\rho} \times e^{(-i2\pi f_s d)} \right) \end{aligned} \quad (11)$$

in the far-field. It can be interpreted through above Eq. (11) and also through our understanding of the convolution of symmetric and asymmetric functions discussed in Sect. 2, that the vortex core in each polarization component undergoes different shifts under asymmetric perturbation. The shifts are observed in opposite directions due to the fact that the component vortices that form the vector field singularity are having topological charges of opposite signs [33]. Due to the superposition of two shifted scalar vortex beams, there is a redistribution of phase and intensity of the resultant beam. This leads to the disintegration of vector field polarization singularities.

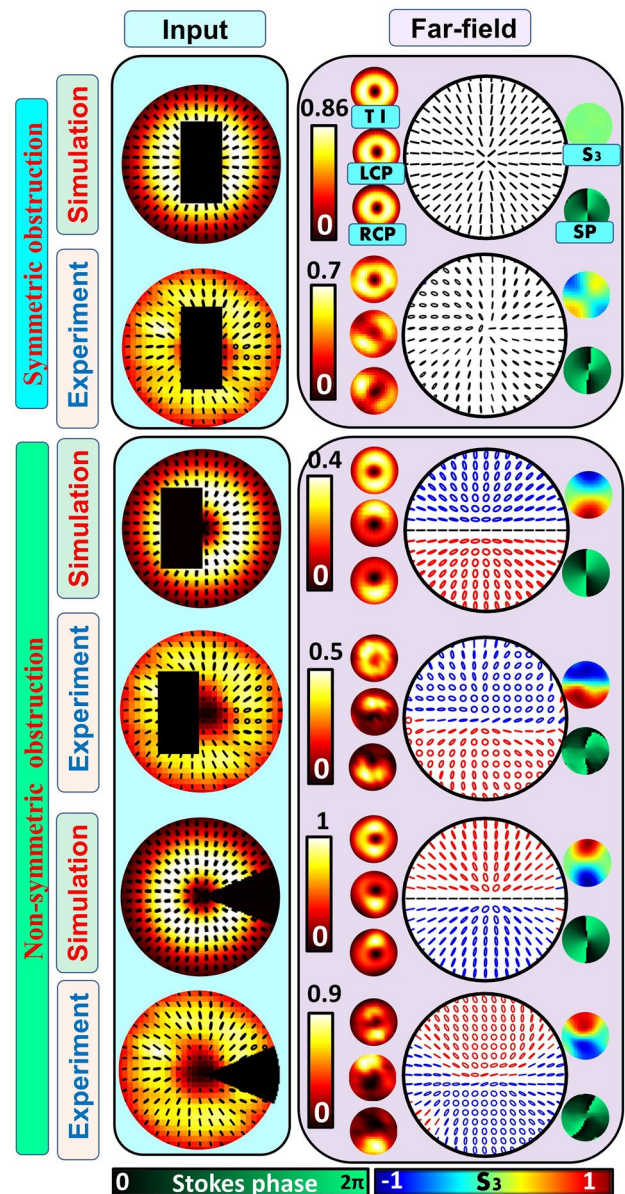


Fig. 5 Splitting of vector vortex beam of index $\eta = 1$ into two C-points on perturbation through non-symmetric obstruction. (TI: total intensity distribution, LCP: left circularly polarized component intensity, RCP: right circularly polarized component intensity, S_3 : stokes parameter S_3 distribution, SP: stokes phase Φ_{12} distribution)

The effect of symmetrical and asymmetrical obstructions on radially polarized V-point beams is depicted in Fig. 5. The total intensity distribution, along with individual component amplitudes, Stokes parameter S_3 , and Stokes phase Φ_{12} distributions are shown at the far-field for all the cases considered. For the first case, a radially polarized beam is obstructed by a rectangular obstacle placed symmetrically about the optic axis. The donut intensity pattern and radial polarization distribution are regained in the far-field. The distribution of Stokes parameter S_3 shows values almost

approaching zero assuring the formation of V-points. The rectangle is then moved off-axis to make the obstruction non-symmetric about the singular point. This non-symmetric perturbation leads to the splitting of the radially polarized V-point into two C-points (lemons) with opposite helicities. The distribution of Stokes parameter S_3 confirms this splitting. The resultant intensity, however, seems to be self-healing, but the polarization distribution and Stokes phase show there is no complete healing. The overall azimuth distribution in the polarization pattern is regained, which indicates that there is partial polarization healing. However, the ellipticity distribution in the beam suggests that the healing process is incomplete.

A similar observation is also made for sectorial type obstruction. Therefore, it can be concluded that symmetry plays a major role in the self-healing of a vector vortex beam. The perturbation of the V-points with asymmetric obstacles may also be used as a generation technique for producing C-points from V-points. In this process of splitting, the index and helicity of the beam remain conserved [34, 35]. For the V-points with index $\eta = \pm 1$, it splits into two C-points with indices $\pm 1/2$ each. These C-points are orthogonal to each other, i.e., one of them is a right-handed lemon and another one is a left-handed lemon, in order to conserve the index and helicity. L-line is also observed in between the two oppositely handed C-points. On similar lines, a V-point of index $\eta = \pm 2$ splits into four C-points each with indices $\pm 1/2$, as shown in Fig. 6 through simulations and experimentation. Therefore, splitting of the input vector vortex beams under asymmetric perturbation can also be used for the detection of indices.

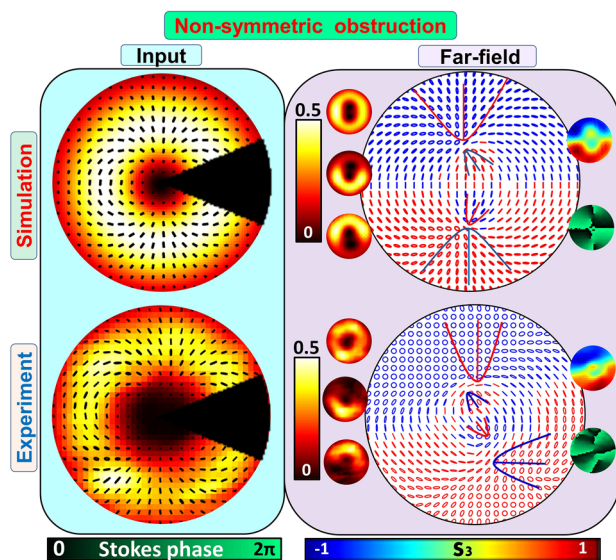


Fig. 6 Splitting of vector vortex beam of index $\eta = 2$ into two C-points on perturbation through the non-symmetric obstruction

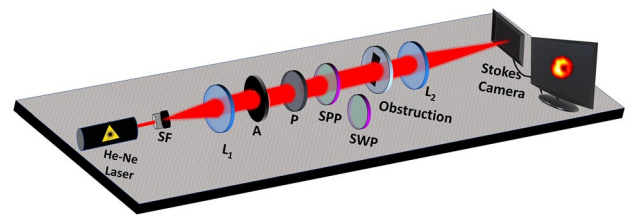


Fig. 7 Schematic of the experimental setup used for examining the effect of symmetric and non-symmetric obstruction on the healing of the intensity distribution of vector vortex beams. *SF*: spatial filter assembly, *L*₁, *L*₂: lens, *A*: aperture, *P*: polarizer, *SPP*: spiral phase plate, *SWP*: spatially varying wave plate

4 Experimental setup

The setup shown in Fig. 7 is used to investigate the role of symmetry in the self-healing of optical singularities. A He-Ne laser beam of wavelength 632.8 nm is spatially filtered and collimated, and a circularly symmetric beam of diameter 2 mm is obtained. The spatial filtering of the beam by the pinhole provides the central lobe of the Airy beam that closely resembles a Gaussian beam. The size of this Gaussian beam is controlled by a circular aperture. The vortex component is then imparted into the beam with the help of a spiral phase plate (SPP). To generate a radially polarized input beam, the SPP is replaced by a spatially varying wave plate (SWP).

The test input beam is then passed through various amplitude obstructions. A centrally obstructing triangular and rectangular obstructions are used to study the symmetric perturbation of optical vortices. On the other hand, a shifted rectangular obstacle and sectorial obstacle are considered to study the effect of asymmetric perturbation on a vector vortex beam. The far-field intensity distribution and Stokes parameters are recorded using the Stokes Camera (Salsa full Stokes polarization imaging camera, 1040 × 1040 pixels, BOSSA Nova, USA) placed at the back focal plane of the lens *L*₂. All the above discussions about the role of symmetry in the self-healing of optical singularities are verified experimentally through this setup (Fig. 7). The experimental results are shown along with the simulation results in Figs. 3, 4, 5, and 6.

5 Conclusions

Self-healing of scalar and vector vortex beams is studied taking symmetry of obstruction into consideration. It is observed that the scalar, as well as vector vortex beams, completely heal themselves under the symmetric obstruction. For an asymmetric obstruction, the vector vortex

beams split into multiple stable C-points following index, and helicity conservation rules. This is due to the shifts in different directions in the RCP and LCP components of the vector vortex beam caused due to asymmetric perturbation. Here, polarization healing is an important condition to verify the healing process. S_3 and Φ_{12} distributions are plotted to clearly notice the splitting. The breaking of V-points due to asymmetry can be used to generate C-points as well as to detect the index of V-points.

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Data Availability The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

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